

Information

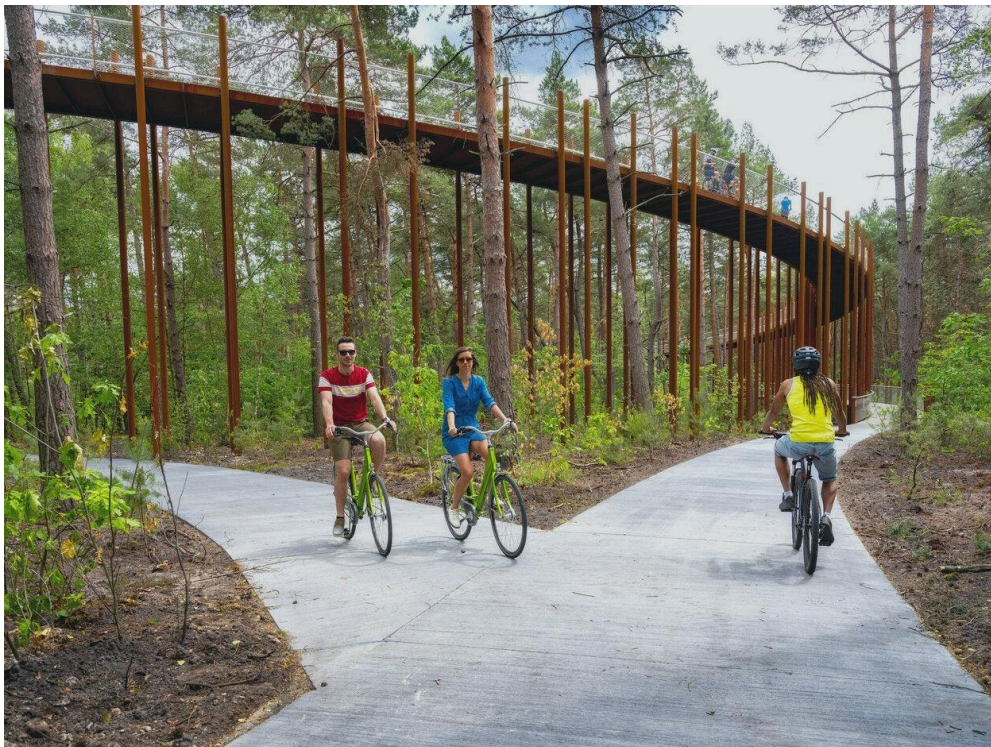
Group Project Information

The course evaluation depends for a large part on a group project that needs to be submitted in groups of 4 students. During the group project, you will engage in a data science project in which you are free to explore your own topics of interests using a dataset provided by the teaching team.

Case Study: Cycling Traffic Data in Flanders

Background

In Flanders (the Dutch-speaking region of Belgium), the **Agentschap Wegen en Verkeer (AWV)** is the Flemish government agency responsible for managing regional roads, highways, and most major cycling infrastructure. AWV's mission is to ensure safe, efficient, and sustainable mobility across the region, which includes planning, constructing, and maintaining thousands of kilometres of bicycle paths, roadways, and bridges.



Source: Cycling through the Trees in Bosland, Limburg

As part of its monitoring and policy efforts, AWV collects **automatic bicycle count data**, known as *fietstellingen*, which records how many bicycles pass at specific counting stations along these cycling routes. This information helps AWV:

- Monitor cycling traffic trends and infrastructure usage
- Evaluate the impact of new cycling paths or road changes
- Plan maintenance and improvements efficiently
- Support policy decisions aimed at promoting cycling as a sustainable transport mode



Source: Agentschap Wegen en Verkeer

Dataset

The *fietstellingen* data is publicly available as open data and is updated monthly in CSV format. It is organized into **three complementary datasets**:

1. Counts (e.g., data-2019-08.csv)

- These monthly CSV files contain the actual automatic bicycle counts recorded by EcoCounter sensors.
- Columns include:

Column	Type	Description
site ID	int	Foreign key linking to the site where the count was recorded
richting	text	Direction of traffic: IN, OUT, or IN/OUT
type	text	Type of count: e.g., FIETSERS (cyclists), VOETGANGERS (pedestrians), AUTOS (cars), etc.
van	datetime	Start time of the counting interval
tot	datetime	End time of the counting interval

aantal	int	Number of items counted during the interval
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Remark: The *type* variable indicates what was detected by the sensor. Possible values include:

- VOETGANGERS (pedestrians)
- FIETTERS (cyclists)
- PAARDEN (horses)
- AUTOS (cars)
- BUSSEN (buses)
- MINIBUSSEN (minibuses)
- MOTORFIETSEN (motorcycles)
- KAYAKS
- NIET GEDEFINIEERD (undefined)

2. Directions (richtingen.csv)

- Provides metadata for the traffic directions recorded at each counting station.
- Columns include:

Column	Type	Description
site ID	int	Links to the counting site in sites.csv
richting	text	Direction: IN, OUT, or IN/OUT
naam	text	Text description of the direction (e.g., "towards city center")

3. Sites (sites.csv)

- Provides metadata about the counting stations themselves.
- Columns include:

Column	Type	Description
site ID	int	Primary key
site nr	int	EcoCounter device ID of site
long	float	WGS84 longitude
lat	float	WGS84 latitude

naam	text	Name of the site
domein	text	Name of the domain where the site is located
wegnr	text	Road number of the site
district	text	Road district number
gemeente	text	Municipality of the site
interval	int	Counting interval in minutes
datum_van	date	Date the sensor was installed

Together, these three datasets allow for a **comprehensive analysis of cycling (or maybe other) behaviour**, combining temporal, directional, and spatial perspectives.

Note: External data (weather, events, infrastructure projects) can be combined for advanced analysis in this project. Think out of the box and impress us!

Where to find the data?

AWV makes this data publicly available here:

- Dataset catalogue:
<https://www.vlaanderen.be/datavindplaats/catalogus/fietstellingen-awv>
- Direct CSV files:
<https://opendata.apps.mow.vlaanderen.be/fietstellingen/index.html>

Project Proposal

The first step in your project is to write a project proposal of 1-3 pages in which you explain what you will do with this data and how you will do this (tools/packages you will use). The teaching team will provide you with feedback and suggestions based on your proposal.

!!Important reminder: The deadline to the project proposal April 3rd, 2025 (23h59 BE time).

Please make sure to follow this deadline! Late submission will automatically result in a NA (Not Attended) score!!